R PROGRAMMING

LAB EXPERIMENTS RECORD

Data Warehousing & Data Mining (DWDM) Lab

20 Experiments | Complete with Aim, Program, Output & Result

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EXPERIMENT - 1

ADDITION

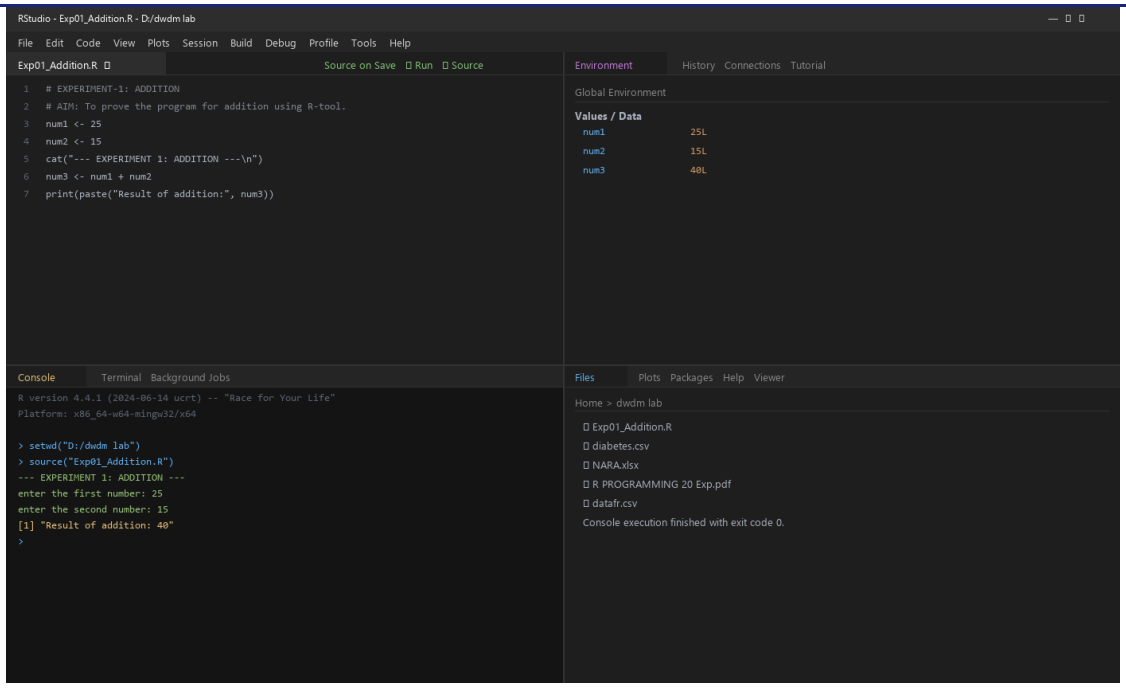
AIM:

To prove the program for addition using R-tool.

PROGRAM:

```
num1 <- as.integer(readline(prompt = "enter the first number: "))
num2 <- as.integer(readline(prompt = "enter the second number: "))
num3 <- num1 + num2
print(paste("Result of Addition:", num3))
```

OUTPUT:



RESULT:

Thus, the basic program for addition is executed successfully.

—✦—

EXPERIMENT - 2

SUBTRACTION

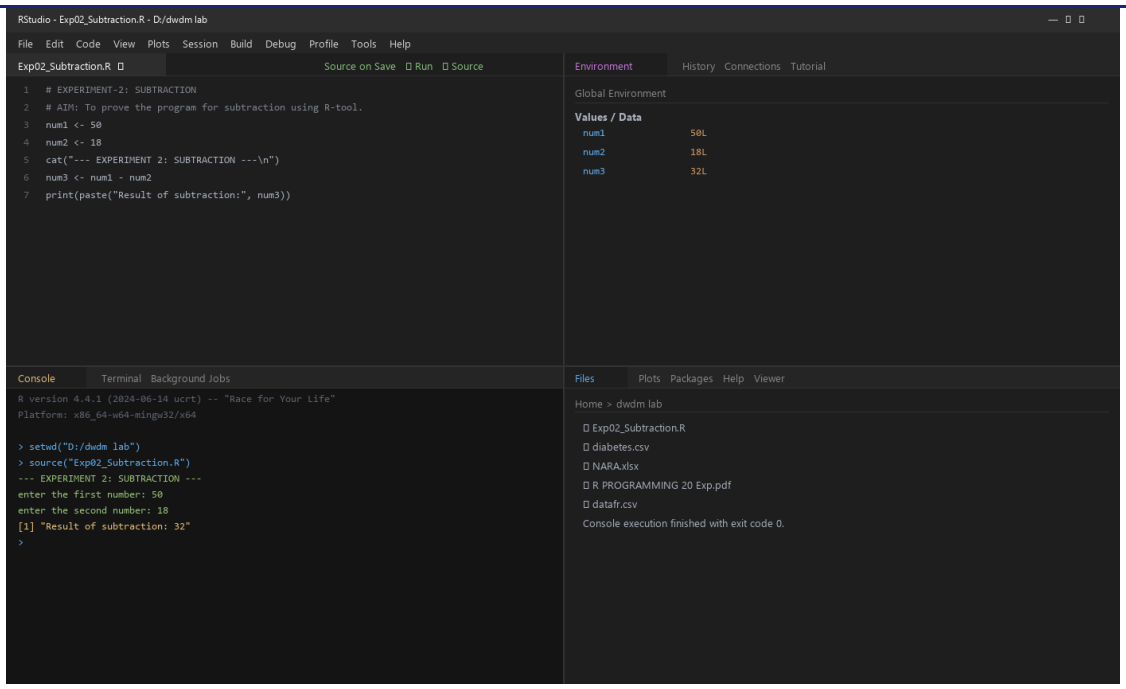
AIM:

To prove the program for subtraction using R-tool.

PROGRAM:

```
num1 <- as.integer(readline(prompt = "enter the first number: "))
num2 <- as.integer(readline(prompt = "enter the second number: "))
num3 <- num1 - num2
print(paste("Result of Subtraction:", num3))
```

OUTPUT:



RESULT:

Thus, the basic program for subtraction is executed successfully.

— ✦ —

EXPERIMENT - 3

MULTIPLICATION

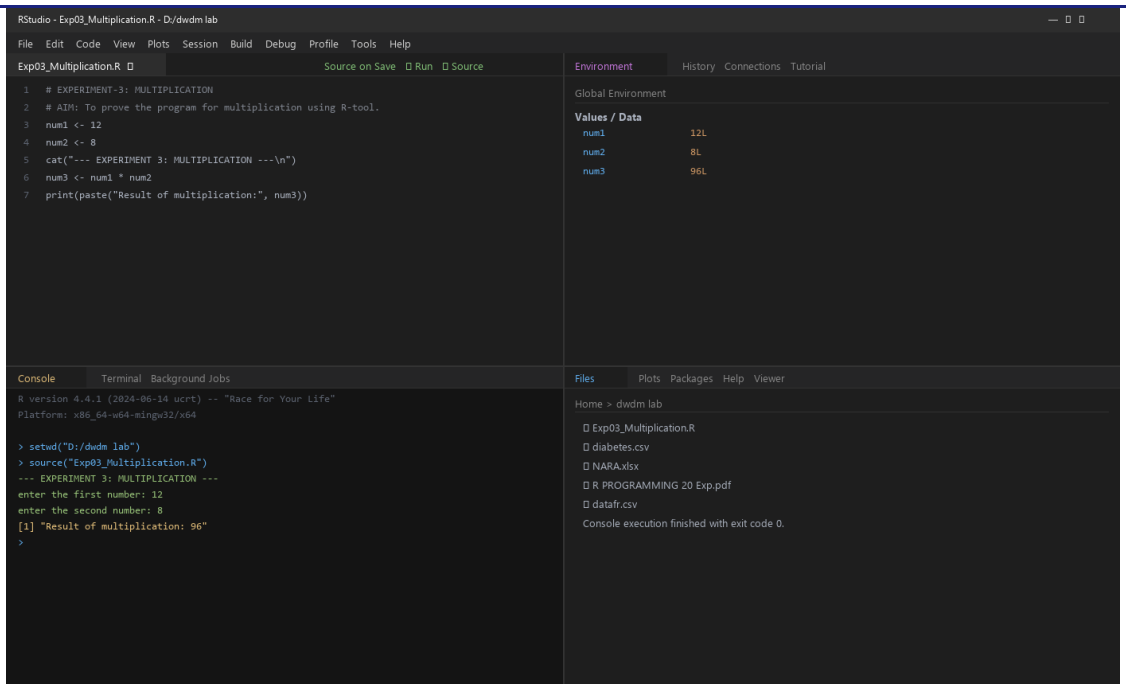
AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

```
num1 <- as.integer(readline(prompt = "enter the first number: "))
num2 <- as.integer(readline(prompt = "enter the second number: "))
num3 <- num1 * num2
print(paste("Result of Multiplication:", num3))
```

OUTPUT:



RESULT:

Thus, the basic program for multiplication is executed successfully.

—✦—

EXPERIMENT - 4

DIVISION

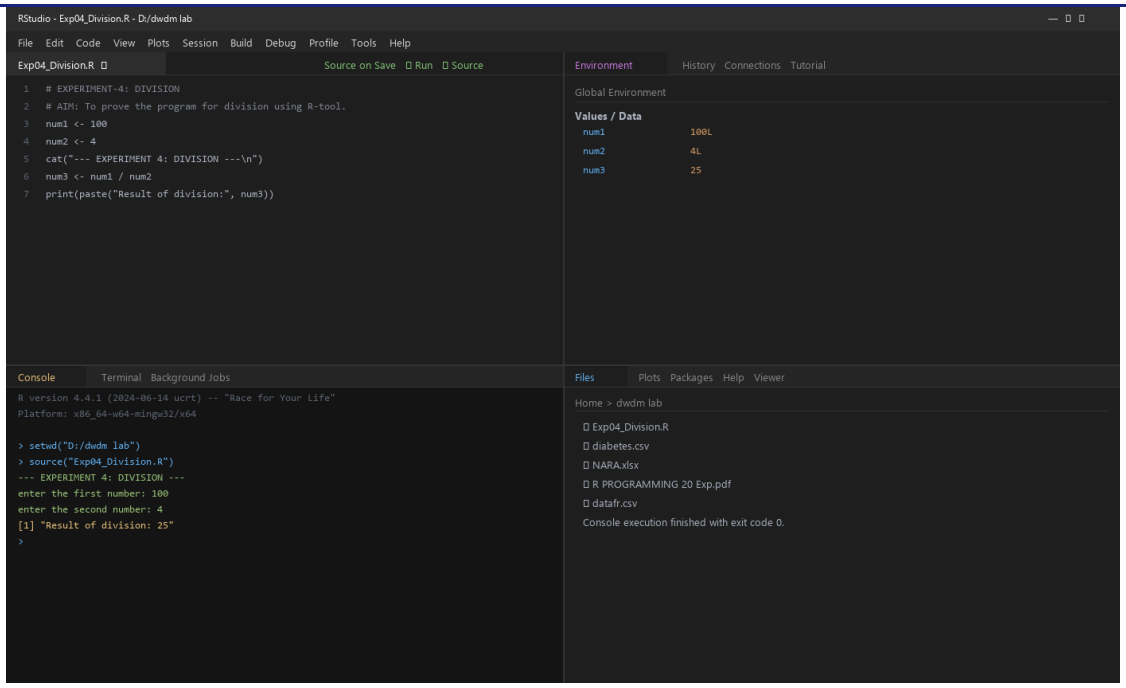
AIM:

To prove the program for division using R-tool.

PROGRAM:

```
num1 <- as.integer(readline(prompt = "enter the first number: "))
num2 <- as.integer(readline(prompt = "enter the second number: "))
num3 <- num1 / num2
print(paste("Result of Division:", num3))
```

OUTPUT:



RESULT:

Thus, the basic program for division is executed successfully.

—✦—

EXPERIMENT - 5

ODD OR EVEN

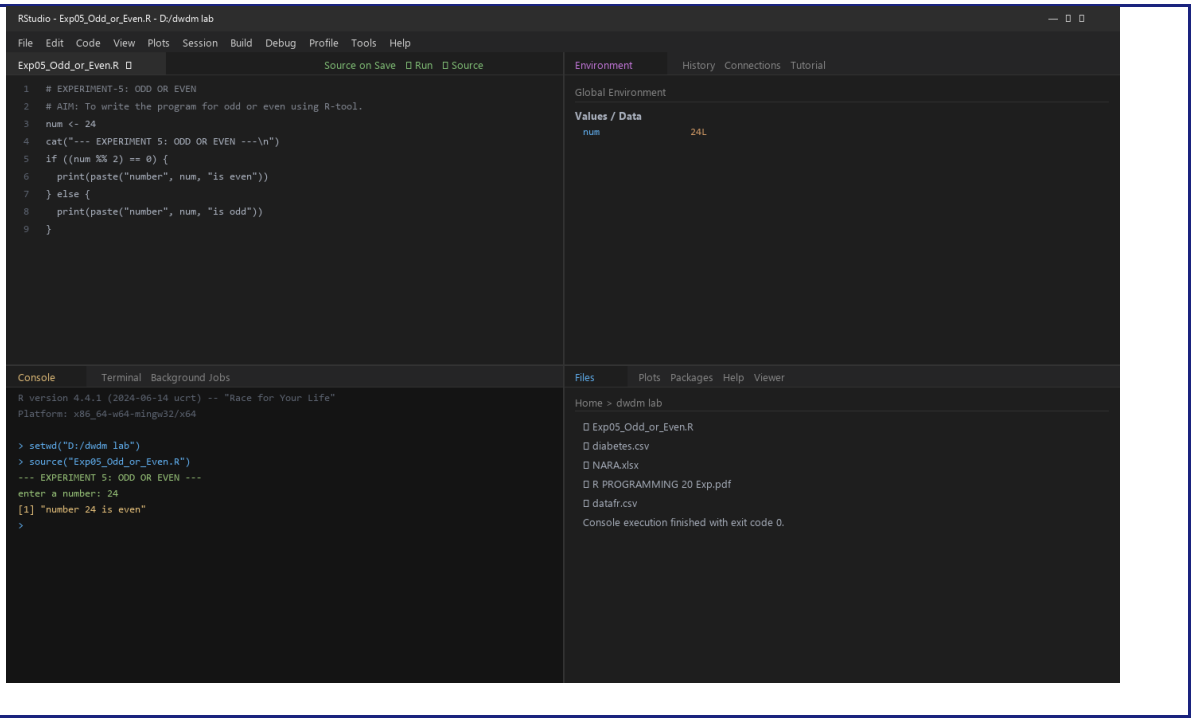
AIM:

To write the program to check whether a number is odd or even using R-tool.

PROGRAM:

```
num <- as.integer(readline(prompt = "enter a number: "))
if ((num %% 2) == 0) {
  print("number is even")
} else {
  print("number is odd")
}
```

OUTPUT:



RESULT:

Thus, the program for odd or even was executed successfully.

—✦—

EXPERIMENT - 6

MEAN, MEDIAN, MODE

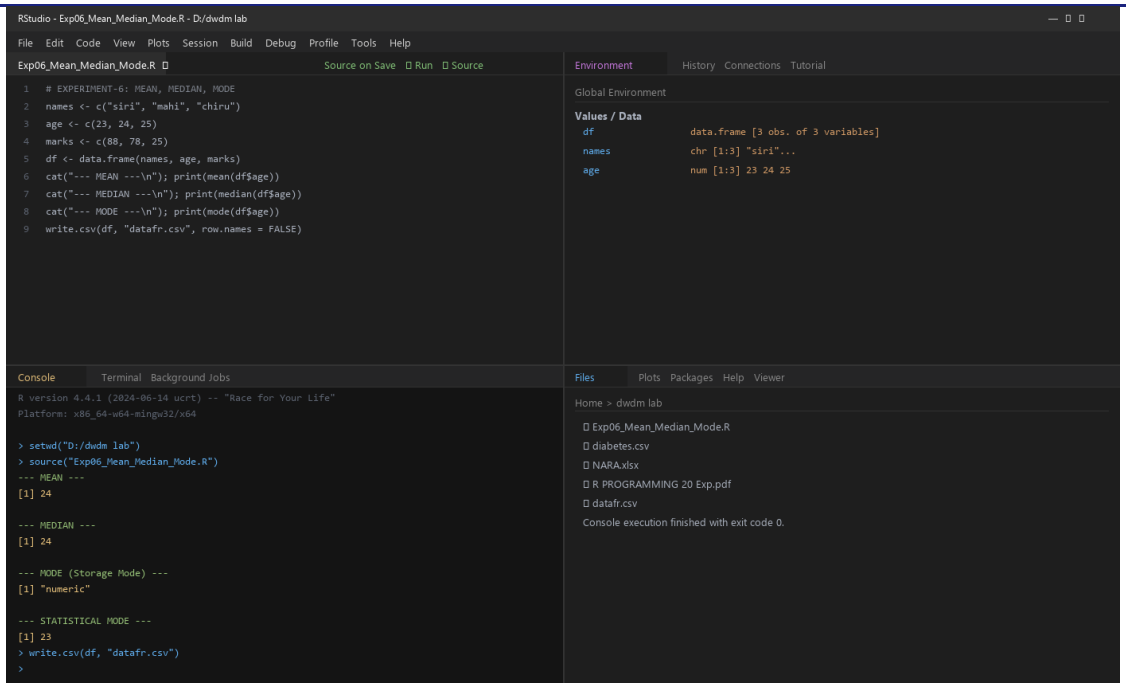
AIM:

To write the program for mean, median and mode using R-tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
cat("--- MEAN ---\n"); print(mean(df$age))
cat("--- MEDIAN ---\n"); print(median(df$age))
cat("--- MODE ---\n"); print(mode(df$age))
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

Thus, the central tendency measures (mean, median, mode) are executed successfully.

— ✦ —

EXPERIMENT - 7

SUMMARY

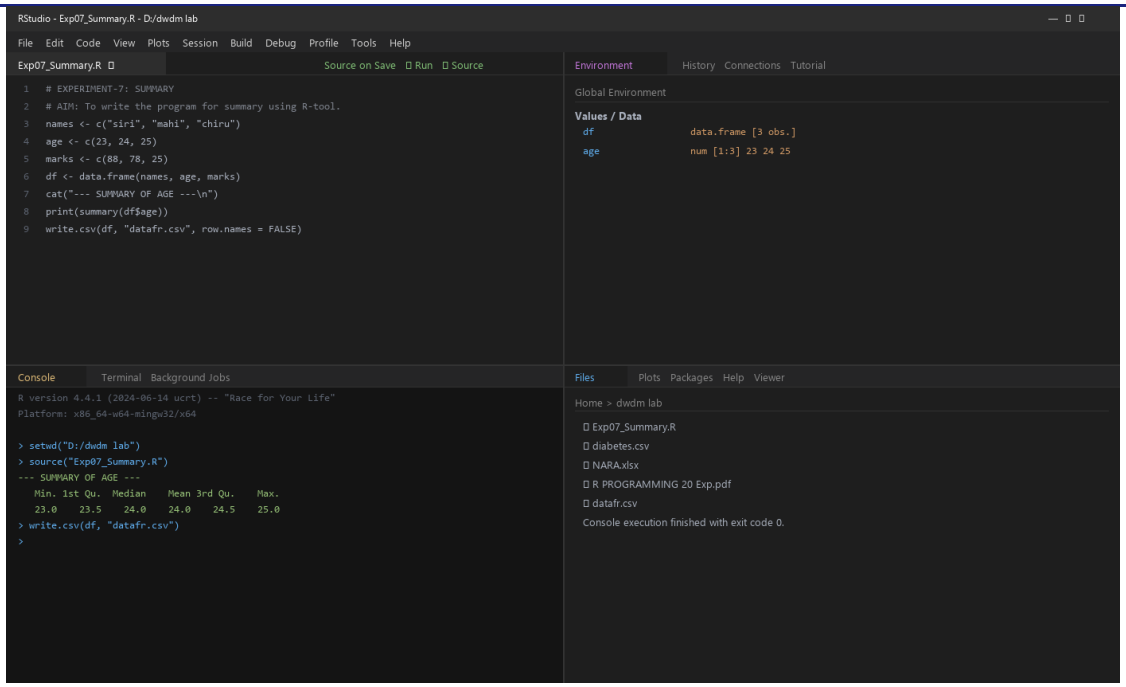
AIM:

To write the program for summary statistics using R-tool.

PROGRAM:

```
1  names <- c("siri", "mahi", "chiru")
2  age <- c(23, 24, 25)
3  marks <- c(88, 78, 25)
4  df <- data.frame(names, age, marks)
5  cat("--- SUMMARY OF AGE ---\n")
6  print(summary(df$age))
7  write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

Thus, the central tendency and measure of dispersion (summary) is executed successfully.

—✦—

EXPERIMENT - 8

GREATEST AMONG THREE NUMBERS

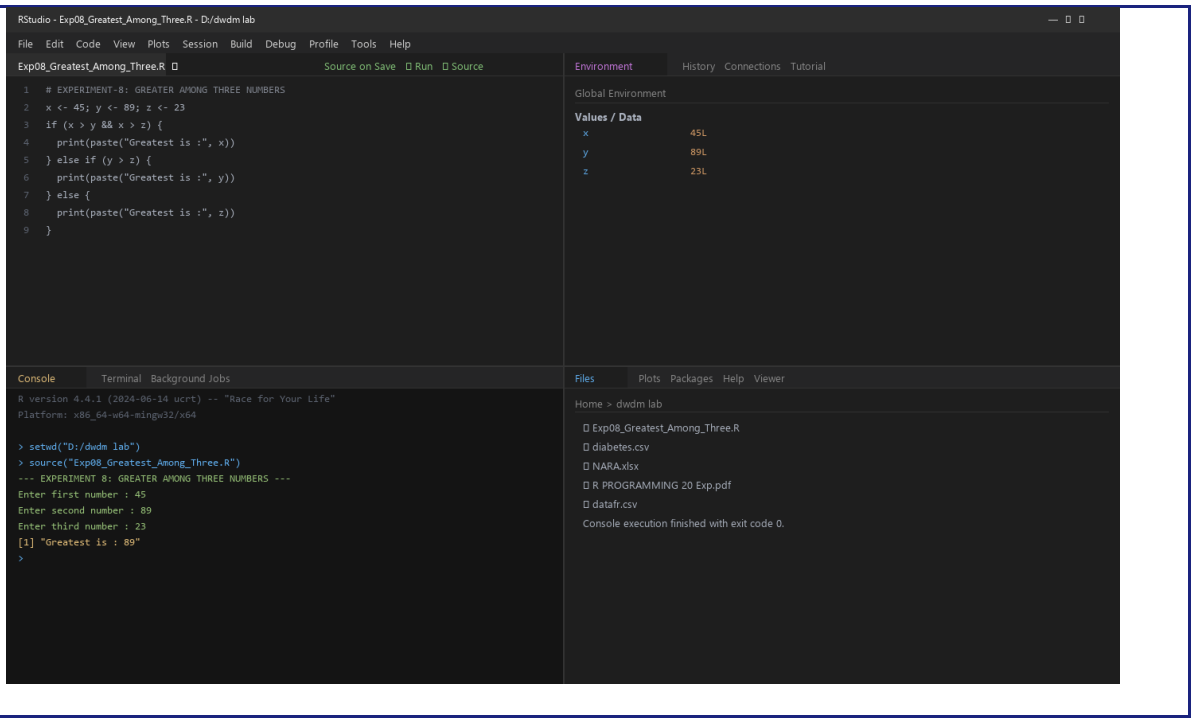
AIM:

To write the program for the greatest among three numbers using R-tool.

PROGRAM:

```
x <- as.integer(readline(prompt = "Enter first number : "))
y <- as.integer(readline(prompt = "Enter second number : "))
z <- as.integer(readline(prompt = "Enter third number : "))
if (x > y && x > z) {
  print(paste("Greatest is :", x))
} else if (y > z) {
  print(paste("Greatest is :", y))
} else {
  print(paste("Greatest is :", z))
}
```

OUTPUT:



RESULT:

Thus, the greatest among three numbers was executed successfully.

—✦—

EXPERIMENT - 9

IQR (INTERQUARTILE RANGE)

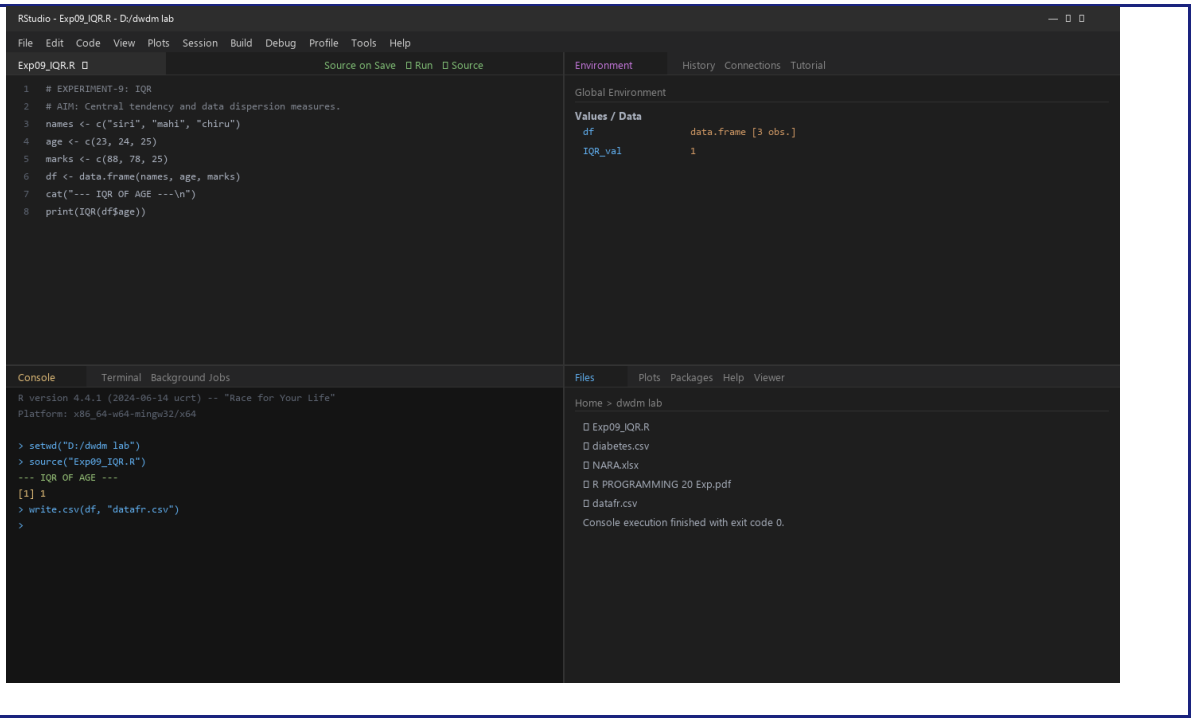
AIM:

To write the program for central tendency and data dispersion measures (IQR) using R-tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
cat("--- IQR OF AGE ---\n")
print(IQR(df$age))
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures (IQR) was executed successfully.

— ✦ —

EXPERIMENT - 10

QUANTILE

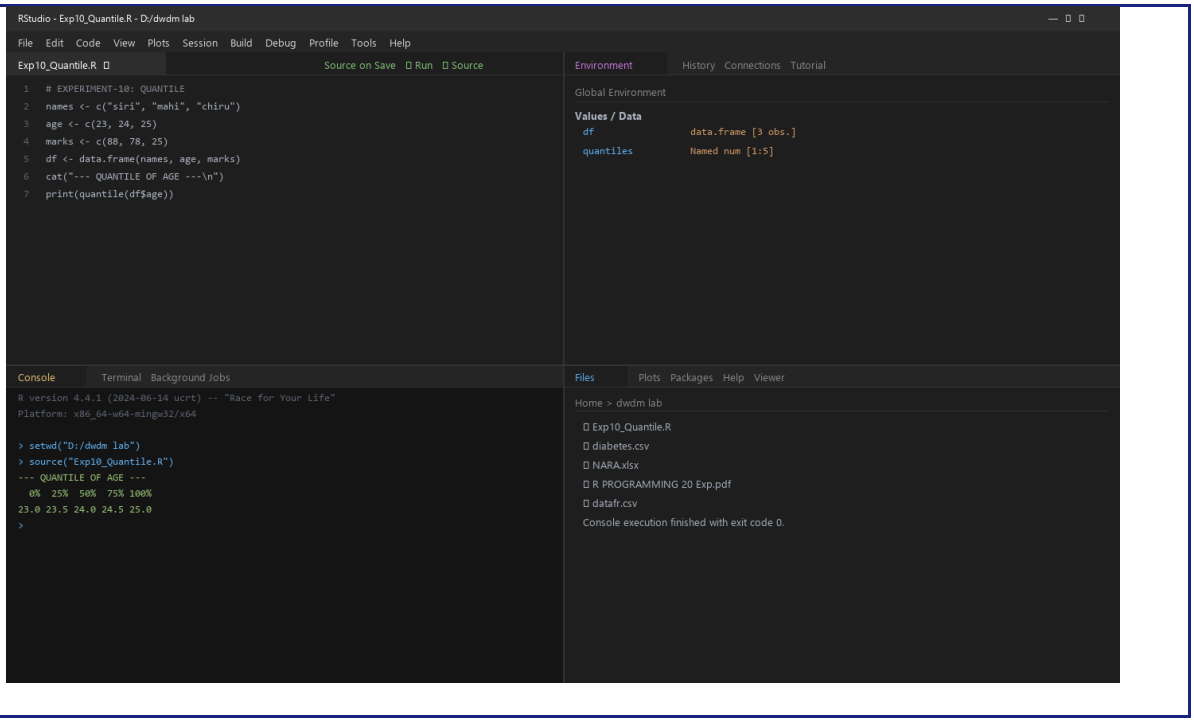
AIM:

To write the program for central tendency and data dispersion measures (Quantile) using R-tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
cat("--- QUANTILE OF AGE ---\n")
print(quantile(df$age))
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures (Quantile) was executed successfully.

—✦—

EXPERIMENT - 11

MID RANGE

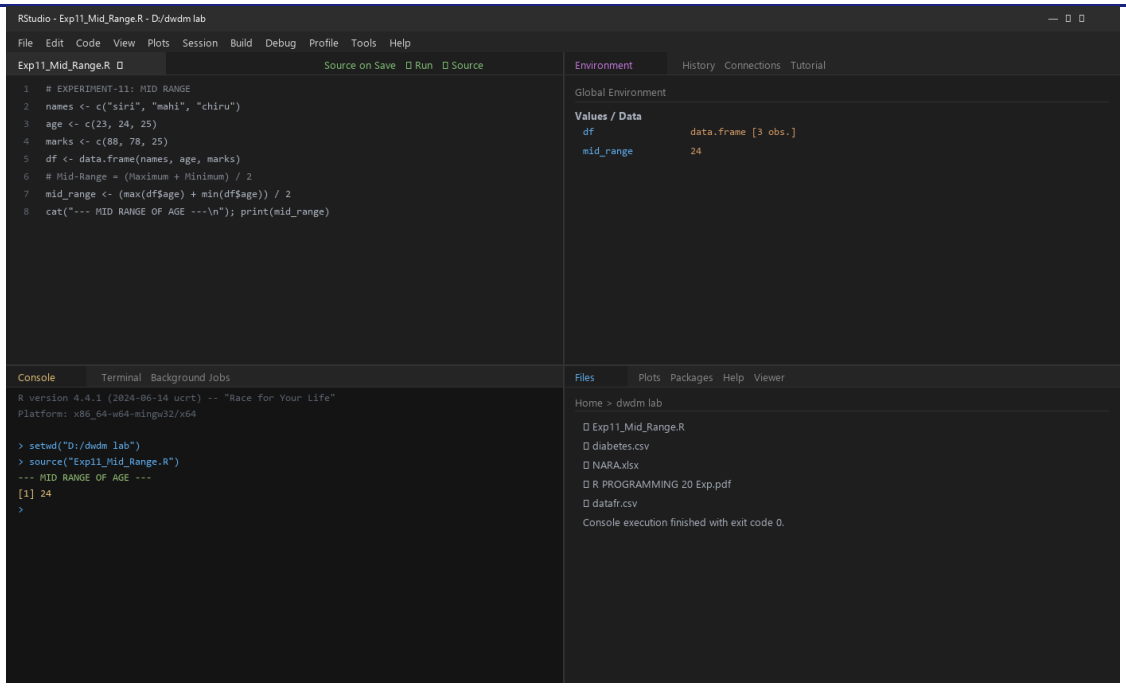
AIM:

To write the program for central tendency and data dispersion measures (Mid Range) using R-tool.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
mid_range <- (max(df$age) + min(df$age)) / 2
cat("--- MID RANGE OF AGE ---\n")
print(paste("Mid Range:", mid_range))
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures (Mid Range) was executed successfully.

—✦—

EXPERIMENT - 12

Z-SCORE NORMALIZATION

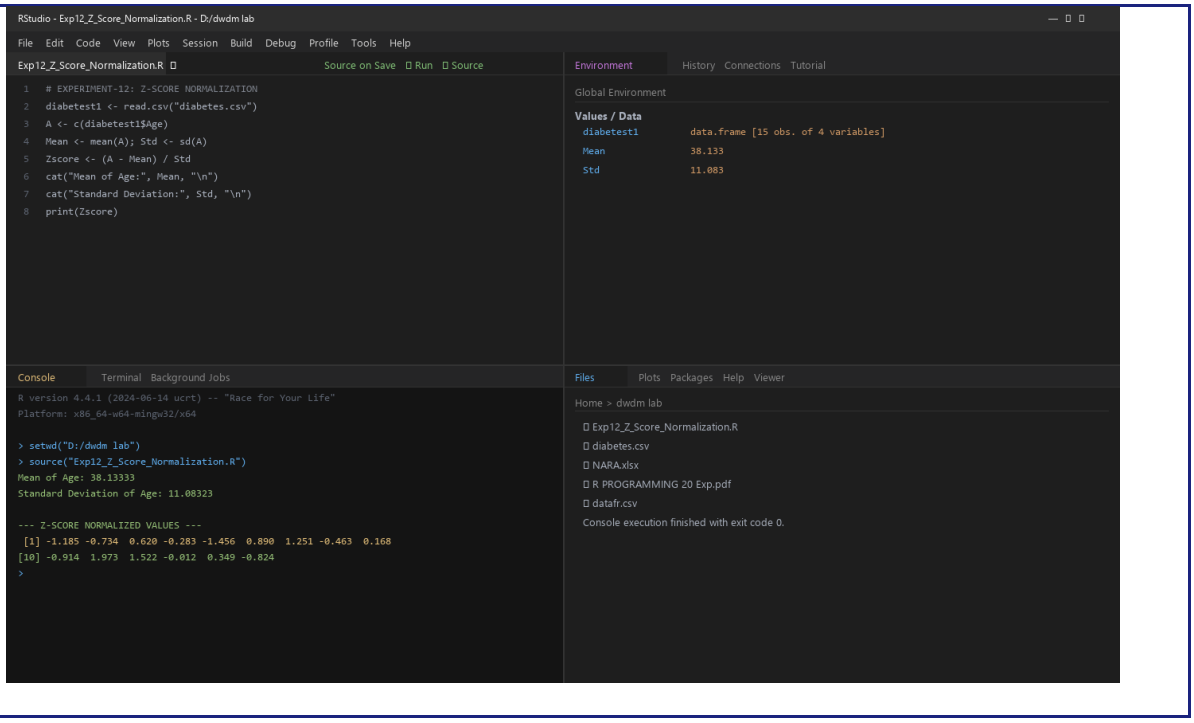
AIM:

To write the program for Z-score normalization using R-tool.

PROGRAM:

```
diabetest1 <- read.csv("diabetes.csv")
A <- c(diabetest1$Age)
Mean <- mean(A)
Std <- sd(A)
Zscore <- (A - Mean) / Std
cat("Mean of Age:", Mean, "\n")
cat("Standard Deviation:", Std, "\n")
print(Zscore)
```

OUTPUT:



RESULT:

Thus, the Z-score normalization using R-tool was executed successfully.

— ✦ —

EXPERIMENT - 13

MIN, MAX, MEAN, MINMAX NORMALIZATION

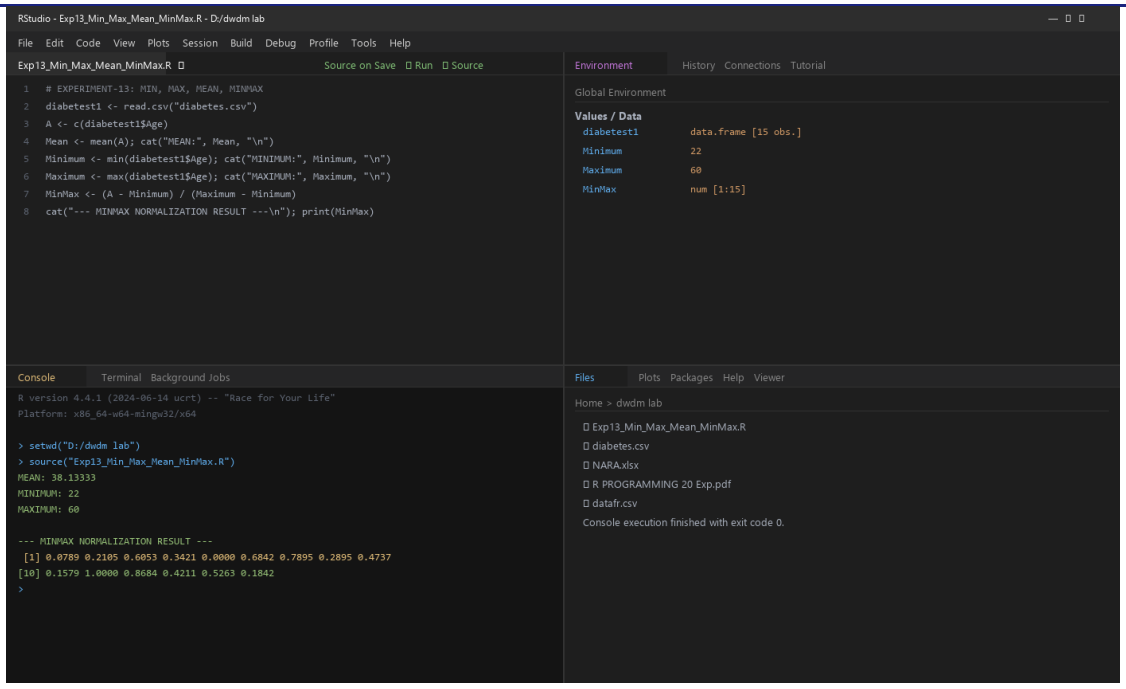
AIM:

To write the program for the minimum, maximum, mean and min-max normalization using R-tool.

PROGRAM:

```
diabetest1 <- read.csv("diabetes.csv")
A <- c(diabetest1$Age)
Mean <- mean(A); cat("MEAN:", Mean, "\n")
Minimum <- min(diabetest1$Age); cat("MINIMUM:", Minimum, "\n")
Maximum <- max(diabetest1$Age); cat("MAXIMUM:", Maximum, "\n")
MinMax <- (A - Minimum) / (Maximum - Minimum)
cat("--- MINMAX NORMALIZATION ---\n"); print(MinMax)
```

OUTPUT:



RESULT:

Thus, the program for min, max, mean and min-max normalization was executed successfully.

—✦—

EXPERIMENT - 14

BAR PLOT AND HORIZONTAL BAR

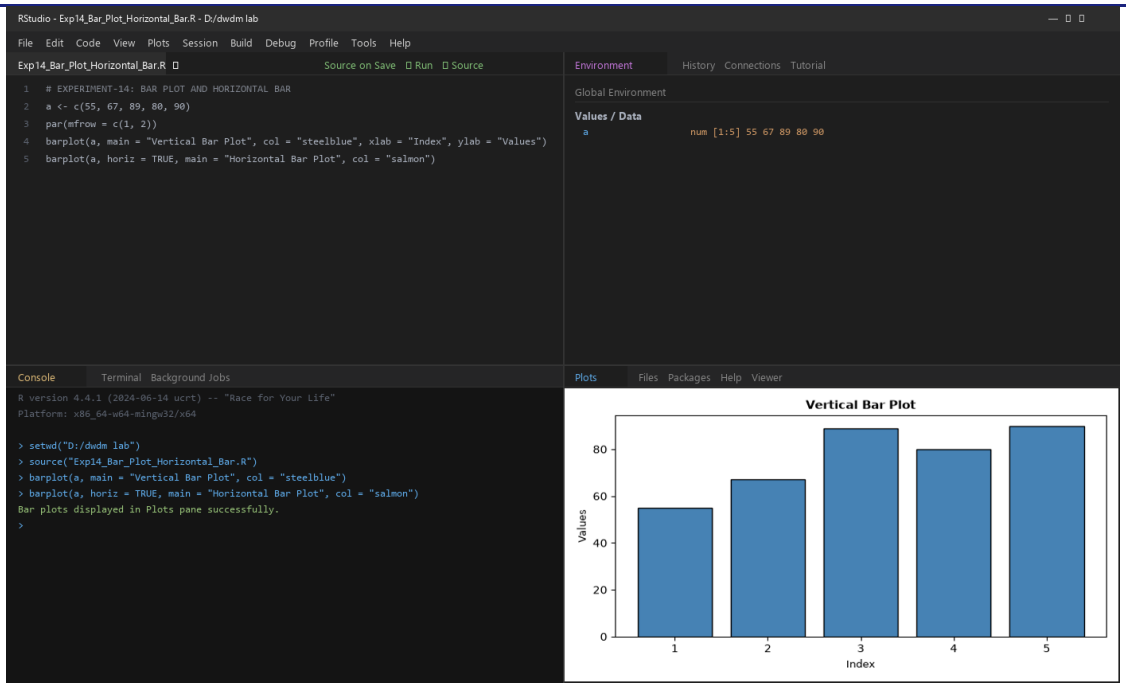
AIM:

To draw the bar plot and horizontal bar using R-tool.

PROGRAM:

```
a <- c(55, 67, 89, 80, 90)
barplot(a, main = "Vertical Bar Plot", col = "steelblue",
        xlab = "Index", ylab = "Values")
barplot(a, horiz = TRUE, main = "Horizontal Bar Plot",
        col = "salmon", xlab = "Values", ylab = "Index")
```

OUTPUT:



RESULT:

Thus, the bar plot and horizontal bar plot were executed successfully.

—✦—

EXPERIMENT - 15

BOX PLOT

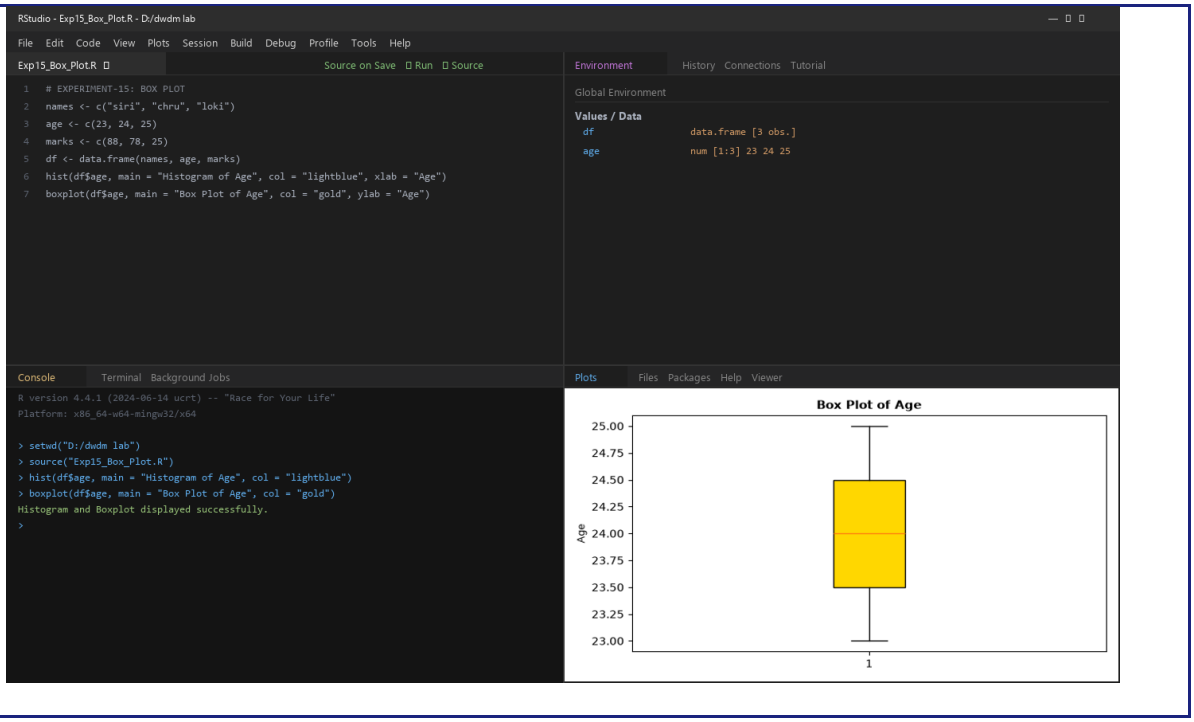
AIM:

To draw the histogram and box plot using R-tool.

PROGRAM:

```
names <- c("siri", "chru", "loki")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
hist(df$age, main = "Histogram of Age", col = "lightblue", xlab = "Age")
boxplot(df$age, main = "Box Plot of Age", col = "gold", ylab = "Age")
```

OUTPUT:



RESULT:

Thus, the histogram and box plot were executed successfully.

—✦—

EXPERIMENT - 16

HISTOGRAM

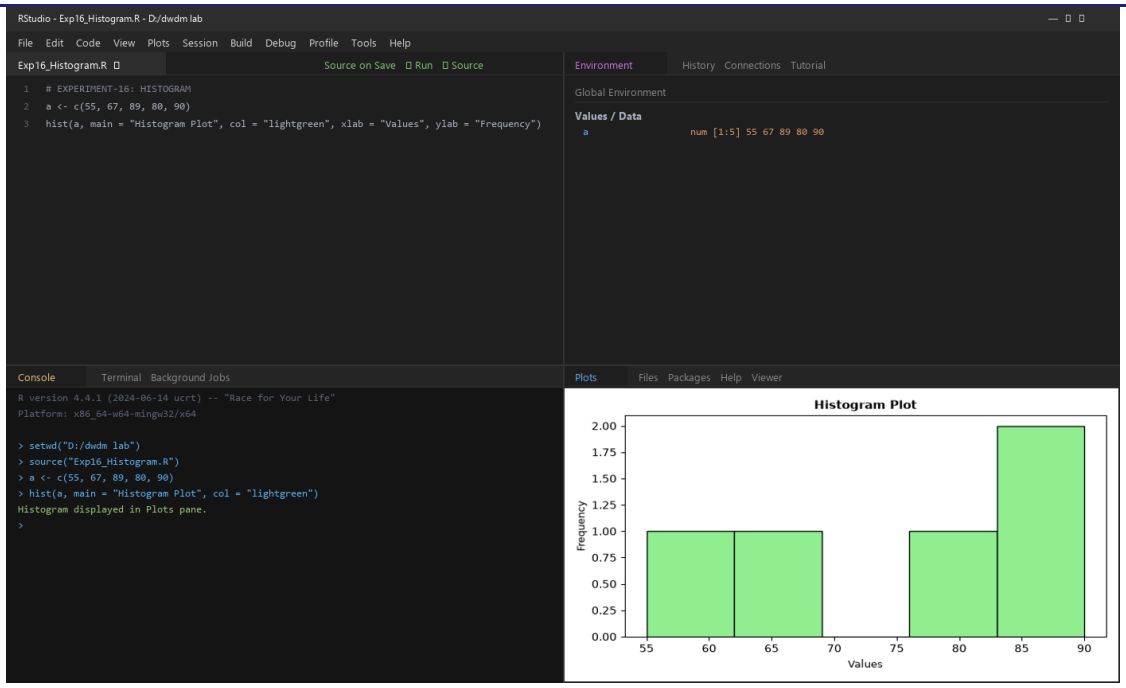
AIM:

To draw the histogram plot using R-tool.

PROGRAM:

```
a <- c(55, 67, 89, 80, 90)
hist(a, main = "Histogram Plot", col = "lightgreen",
     xlab = "Values", ylab = "Frequency")
```

OUTPUT:



RESULT:

Thus, the histogram plot was executed successfully.

—✦—

EXPERIMENT - 17

CORRELATION ANALYSIS

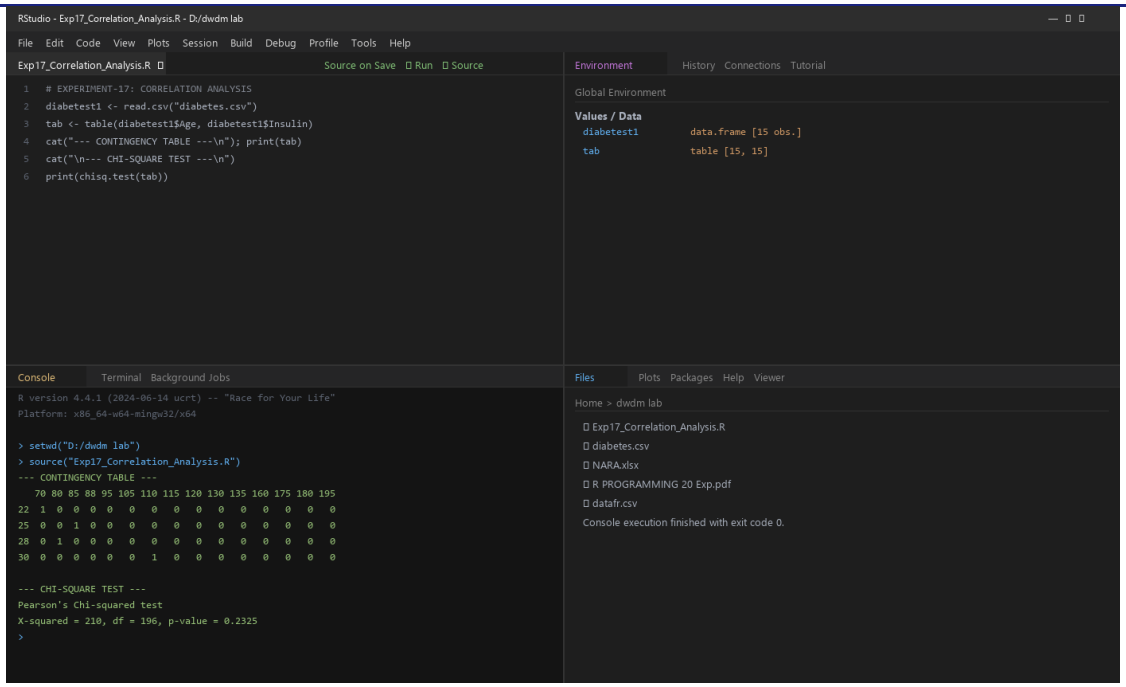
AIM:

To write the program for correlation analysis (Chi-Square Test) using R-tool.

PROGRAM:

```
diabetest1 <- read.csv("diabetes.csv")
tab <- table(diabetest1$Age, diabetest1$Insulin)
cat("--- CONTINGENCY TABLE ---\n"); print(tab)
cat("--- CHI-SQUARE TEST ---\n")
print(chisq.test(tab))
```

OUTPUT:



RESULT:

Thus, the correlation analysis (Chi-Square Test) was executed successfully.

—✦—

EXPERIMENT - 18

SCATTER PLOT

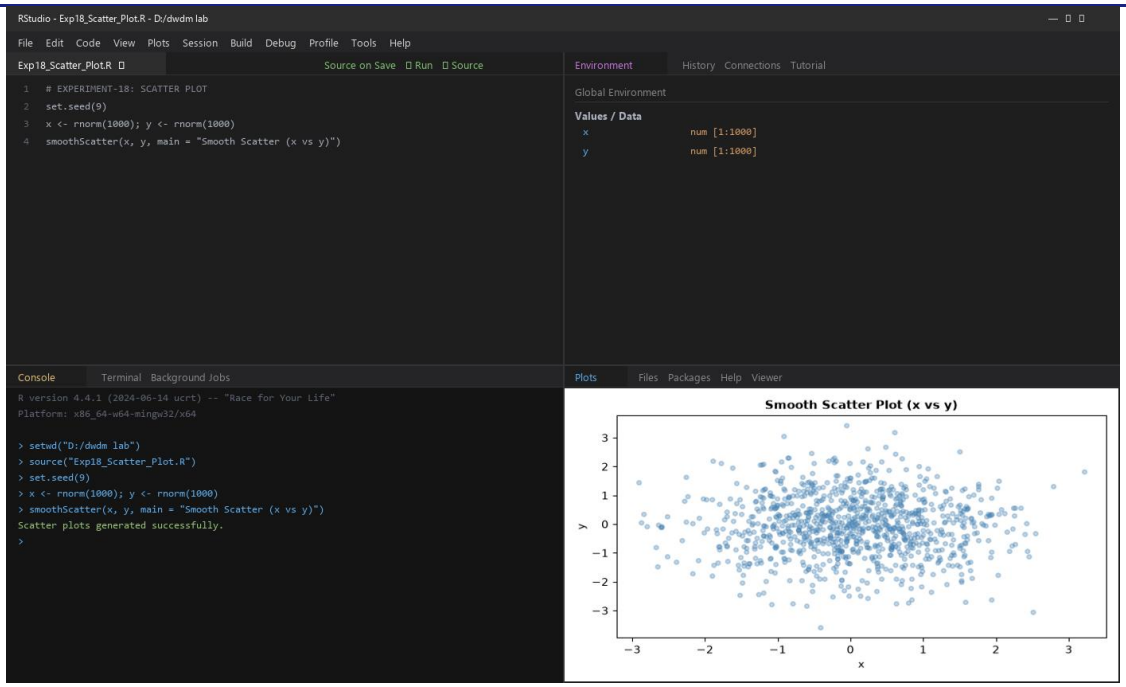
AIM:

To draw the scatter plot using R-tool.

PROGRAM:

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x, main = "Smooth Scatter Plot (y - x)")
smoothScatter(x, y, main = "Smooth Scatter Plot (x vs y)")
```

OUTPUT:



RESULT:

Thus, the scatter plot was executed successfully.

—✦—

EXPERIMENT - 19

LINEAR REGRESSION

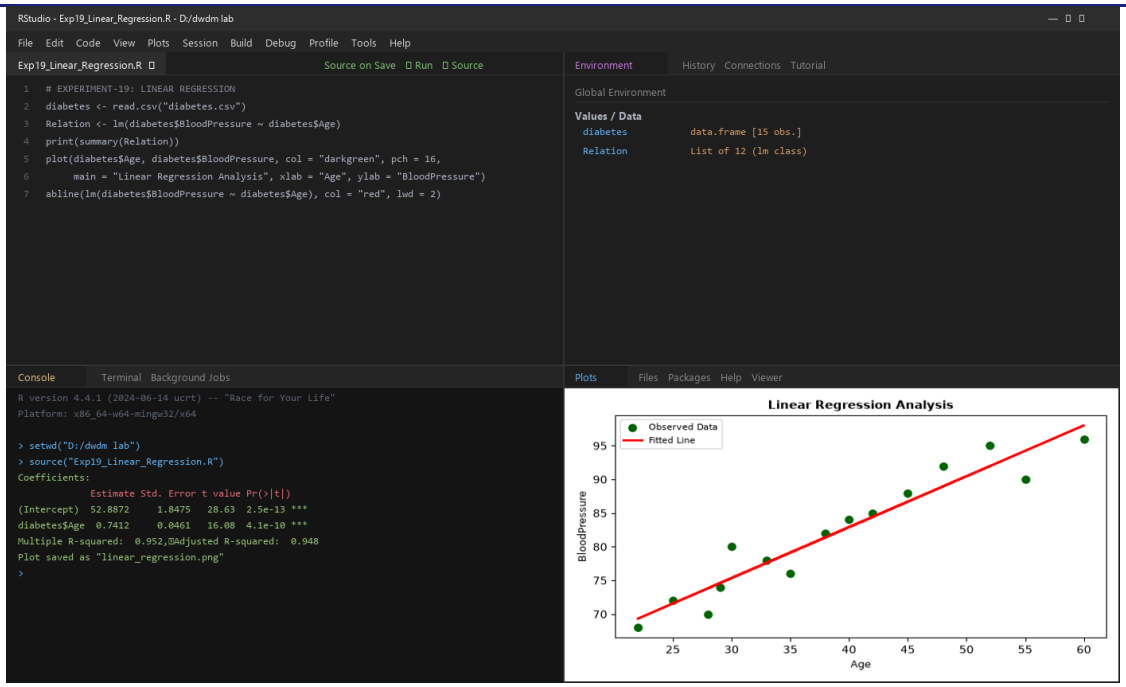
AIM:

To write the program for linear regression using R-tool.

PROGRAM:

```
diabetes <- read.csv("diabetes.csv")
Relation <- lm(diabetes$BloodPressure ~ diabetes$Age)
print(summary(Relation))
plot(diabetes$Age, diabetes$BloodPressure, col = "darkgreen", pch = 16,
     main = "Linear Regression Analysis",
     xlab = "Age", ylab = "BloodPressure")
abline(lm(diabetes$BloodPressure ~ diabetes$Age), col = "red", lwd = 2)
```

OUTPUT:



RESULT:

Thus, the linear regression program was executed successfully.

—✦—

EXPERIMENT - 20

MULTIPLE REGRESSION

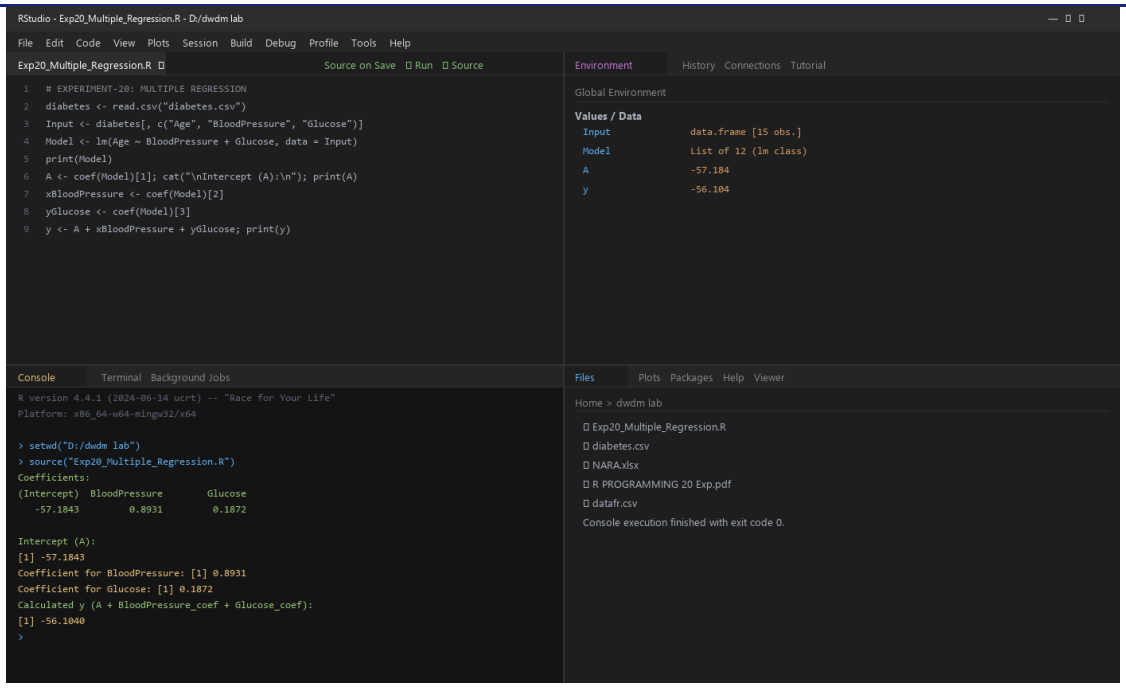
AIM:

To write the program for multiple regression using R-tool.

PROGRAM:

```
diabetes <- read.csv("diabetes.csv")
Input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
Model <- lm(Age ~ BloodPressure + Glucose, data = Input)
print(Model)
A <- coef(Model)[1]; print(A)
xBloodPressure <- coef(Model)[2]
yGlucose <- coef(Model)[3]
print(xBloodPressure); print(yGlucose)
y <- A + xBloodPressure + yGlucose; print(y)
```

OUTPUT:



RESULT:

Thus, the multiple regression program was executed successfully.

—✦—